

Project Definition

Context

The COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) score is a statistically derived measure used to predict a defendant's risk of recidivism, or the likelihood of committing a new crime.

In practice, the COMPAS score is utilized at nearly every stage of the American legal system, with judges, prosecutors, and parole boards relying on it to inform high-stakes decisions.

It promises to replace subjective human intuition with objective, data-driven insights. By providing a standardized risk assessment, it aims to reduce prison overcrowding and improve public safety.

Motivation

One of our individual papers was heavily focused on data fairness. Another one touched a bit in that ballpark while studying different normalization strategies. Because of this we decided to explore further the intersection of model fairness and explainability. These two are important to be able to achieve an unbiased model, and we believe this dataset will allow us to develop better understanding of both.

Data Set

The data contains the records of over 7,000 criminal defendants from Broward County, Florida.

Source

- Originally found in this [kaggle post](#)
- Traced back to [ProPublica's repository](#)

According to their [original study](#):

ProPublica obtained the COMPAS scores for defendants booked between 2013 and 2014. They then mapped these scores to the public criminal records of the same individuals through April 2016 to determine if they actually re-offended (recidivated) within a two-year follow-up period.

The scores were generated by Northpointe's proprietary algorithm, which uses 137

features derived from a defendant's criminal history and their answers to a behavioral and psychological questionnaire.

Limitations

ProPublica's statement about recidivism score reliability:

When a full range of crimes were taken into account — including misdemeanors such as driving with an expired license — the algorithm was somewhat more accurate than a coin flip. Of those deemed likely to re-offend, 61 percent were arrested for any subsequent crimes within two years.

As stated before, the original algorithm uses 137. Trying to estimate COMPAS scores from this dataset may prove to be difficult since we only have 25-26 features. This lack of information may impact our model's performance.

Potencial bias

This dataset is most famous for the racial bias documented in [ProPublica's analysis](#).

It concluded that the algorithm produced significant racial disparities that could not be explained by a defendant's criminal history, age, or gender.

While the tool maintained similar overall accuracy across groups, it exhibited a different error pattern:

- Nearly 2x as likely to falsely flag black defendants as high-risk compared to white defendants
- White defendants were more frequently mislabeled as low-risk.

Even after statistically isolating for prior offenses and recidivism, the study found that:

- Black defendants remained 77% more likely to be predicted as violent recidivists
- Black defendants are also 45% more likely to be predicted to commit any future crime

This suggests the algorithm effectively acted as a proxy for race rather than a neutral predictor of safety.

Our problem

We want to be able to determine a COMPAS score based on a defendant features.
Making this a regression problem.

Initial analysis

Data Loading and Preparation

```
In [ ]: # imports and dataset

import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import skew

# adjust path if needed
path = "compas-scores-raw.csv"
df = pd.read_csv(path)
```

Dataset Overview

```
In [ ]: print(f"Dataset Shape: {df.shape}")
        print(f"Count of Unique Individuals: {df['Person_ID'].nunique()}")
```

Dataset Shape: (60843, 28)
Count of Unique Individuals: 18610

Missing Values

```
In [ ]: print("\nNull or NAN counts:")
        print(df.isnull().sum())
```

```
Null or NAN counts:
Person_ID          0
AssessmentID       0
Case_ID            0
Agency_Text       0
LastName           0
FirstName          0
MiddleName         45219
Sex_Code_Text      0
Ethnic_Code_Text   0
DateOfBirth        0
ScaleSet_ID        0
ScaleSet           0
AssessmentReason   0
Language           0
LegalStatus        0
CustodyStatus      0
MaritalStatus      0
Screening_Date     0
RecSupervisionLevel 0
RecSupervisionLevelText 0
Scale_ID           0
DisplayText        0
RawScore           0
DecileScore        0
ScoreText          45
AssessmentType     0
IsCompleted        0
IsDeleted          0
dtype: int64
```

Variable Types Overview

```
In [ ]: print("\nData types:")
        print(df.dtypes)
```

```
Data types:
Person_ID                int64
AssessmentID             int64
Case_ID                  int64
Agency_Text              str
LastName                 str
FirstName                str
MiddleName               str
Sex_Code_Text            str
Ethnic_Code_Text         str
DateOfBirth              str
ScaleSet_ID              int64
ScaleSet                  str
AssessmentReason          str
Language                  str
LegalStatus               str
CustodyStatus             str
MaritalStatus             str
Screening_Date            str
RecSupervisionLevel       int64
RecSupervisionLevelText   str
Scale_ID                  int64
DisplayText               str
RawScore                  float64
DecileScore               int64
ScoreText                 str
AssessmentType            str
IsCompleted               int64
IsDeleted                 int64
dtype: object
```

Categorical Variable Analysis

```
In [ ]: exclude_vars = ['DateOfBirth', 'MiddleName', 'LastName', 'FirstName', 'Sc
categorical_vars = [col for col in df.select_dtypes(include=['object', 'c
for col in categorical_vars:
    print(f"Col: {col}")
    print(f"Number of categories: {df[col].nunique()}")
    print(f"Percentage:")
    print(df[col].value_counts(normalize=True).mul(100).round(2))
    print()
```

Col: Agency_Text
Number of categories: 4
Percentage:
Agency_Text
PRETRIAL 67.55
Probation 31.32
DRRD 0.93
Broward County 0.20
Name: proportion, dtype: float64

Col: Sex_Code_Text
Number of categories: 2
Percentage:
Sex_Code_Text
Male 78.09
Female 21.91
Name: proportion, dtype: float64

Col: Ethnic_Code_Text
Number of categories: 9
Percentage:
Ethnic_Code_Text
African-American 44.41
Caucasian 35.80
Hispanic 14.37
Other 4.26
Asian 0.53
Native American 0.36
Arabic 0.12
African-Am 0.08
Oriental 0.06
Name: proportion, dtype: float64

Col: ScaleSet
Number of categories: 2
Percentage:
ScaleSet
Risk and Prescreen 96.39
All Scales 3.61
Name: proportion, dtype: float64

Col: AssessmentReason
Number of categories: 1
Percentage:
AssessmentReason
Intake 100.0
Name: proportion, dtype: float64

Col: Language
Number of categories: 2
Percentage:
Language
English 99.59
Spanish 0.41
Name: proportion, dtype: float64

Col: LegalStatus
Number of categories: 7
Percentage:
LegalStatus

Pretrial	61.76
Post Sentence	30.13
Other	7.16
Conditional Release	0.69
Probation Violator	0.21
Parole Violator	0.03
Deferred Sentencing	0.02

Name: proportion, dtype: float64

Col: CustodyStatus
Number of categories: 6
Percentage:
CustodyStatus

Jail Inmate	49.52
Probation	37.75
Pretrial Defendant	12.56
Residential Program	0.11
Parole	0.03
Prison Inmate	0.02

Name: proportion, dtype: float64

Col: MaritalStatus
Number of categories: 7
Percentage:
MaritalStatus

Single	74.17
Married	13.43
Divorced	6.38
Separated	2.96
Significant Other	2.07
Widowed	0.64
Unknown	0.35

Name: proportion, dtype: float64

Col: RecSupervisionLevelText
Number of categories: 4
Percentage:
RecSupervisionLevelText

Low	63.23
Medium	17.22
Medium with Override Consideration	12.86
High	6.69

Name: proportion, dtype: float64

Col: DisplayText
Number of categories: 3
Percentage:
DisplayText

Risk of Violence	33.33
Risk of Recidivism	33.33
Risk of Failure to Appear	33.33

Name: proportion, dtype: float64

Col: ScoreText
Number of categories: 3
Percentage:
ScoreText

Low	68.24
Medium	20.54
High	11.22

Name: proportion, dtype: float64

Col: AssessmentType

Number of categories: 2

Percentage:

AssessmentType

New 92.27

Copy 7.73

Name: proportion, dtype: float64

Numerical Variable Analysis

```
In [ ]: numerical_vars = df.select_dtypes(include=['number']).columns
numerical_vars = [col for col in numerical_vars if 'ID' not in col and co

print("\nNumerical Analysis (Distribution Stats):")
print(df[numerical_vars].describe().round(2))
```

Numerical Analysis (Distribution Stats):

	RecSupervisionLevel	RawScore	DecileScore
count	60843.00	60843.00	60843.00
mean	1.63	5.08	3.57
std	0.94	10.08	2.62
min	1.00	-4.79	-1.00
25%	1.00	-2.09	1.00
50%	1.00	-0.71	3.00
75%	2.00	14.00	5.00
max	4.00	51.00	10.00

```
In [ ]: print("\nSkewness:")
for col in numerical_vars:
    s = skew(df[col].dropna())
    print(f"{col}: {s:.4f}")

plt.figure(figsize=(15, 5))
for i, col in enumerate(numerical_vars, 1):
    plt.subplot(1, 3, i)
    sns.histplot(df[col].dropna(), kde=True, color='skyblue')
    plt.title(f'Distribution of {col}\n(Skewness: {skew(df[col].dropna())}')
plt.tight_layout()
plt.show()
```

Skewness:

RecSupervisionLevel: 1.2727

RawScore: 1.0299

DecileScore: 0.8383

